Assignment 11

Last updated by | Hashim Javed | Sep 2, 2025 at 5:11 PM GMT+5

Docker Assignment

Containerization

- Create Dockerfiles for both the .NET Web API and the Angular frontend.
- Ensure images are built using multi-stage builds to minimize size.
- Add appropriate `.dockerignore` files to exclude unnecessary files.

Application Orchestration

- Define a `docker-compose.yml` that runs both services together.
- Configure networking so the Angular frontend can call the API service.
- Expose services on configurable ports for local access.
- Implement basic health checks for both services.

Configuration Management

- Use environment variables (via `.env` file) for settings such as ports, API base URL, and image version.
- Ensure the frontend can be configured to communicate with the API via environment-specific base URLs.

Image Publishing

- Tag images with both `latest` and semantic version tags.
- Push images to Docker Hub under the correct repository.
- Verify images can be pulled and run on a different machine.

Docker Compose Prod

- Define a `docker-compose.prod.yml` that runs both services together using pushed images from docker hub.

Documentation

- Provide a README with clear setup instructions:
  - Building images
  - Running with Docker Compose
  - Testing connectivity between services
  - Pushing images to Docker Hub
  - Troubleshooting common issues

Acceptance Criteria

- Containers build and run successfully on a clean machine.
- Both services are healthy and accessible via the defined ports.
- The Angular app communicates successfully with the .NET API.

- Docker Hub contains the pushed images with correct tags.
- Documentation is complete and easy to follow.

Submission

- Till September 03, 2025